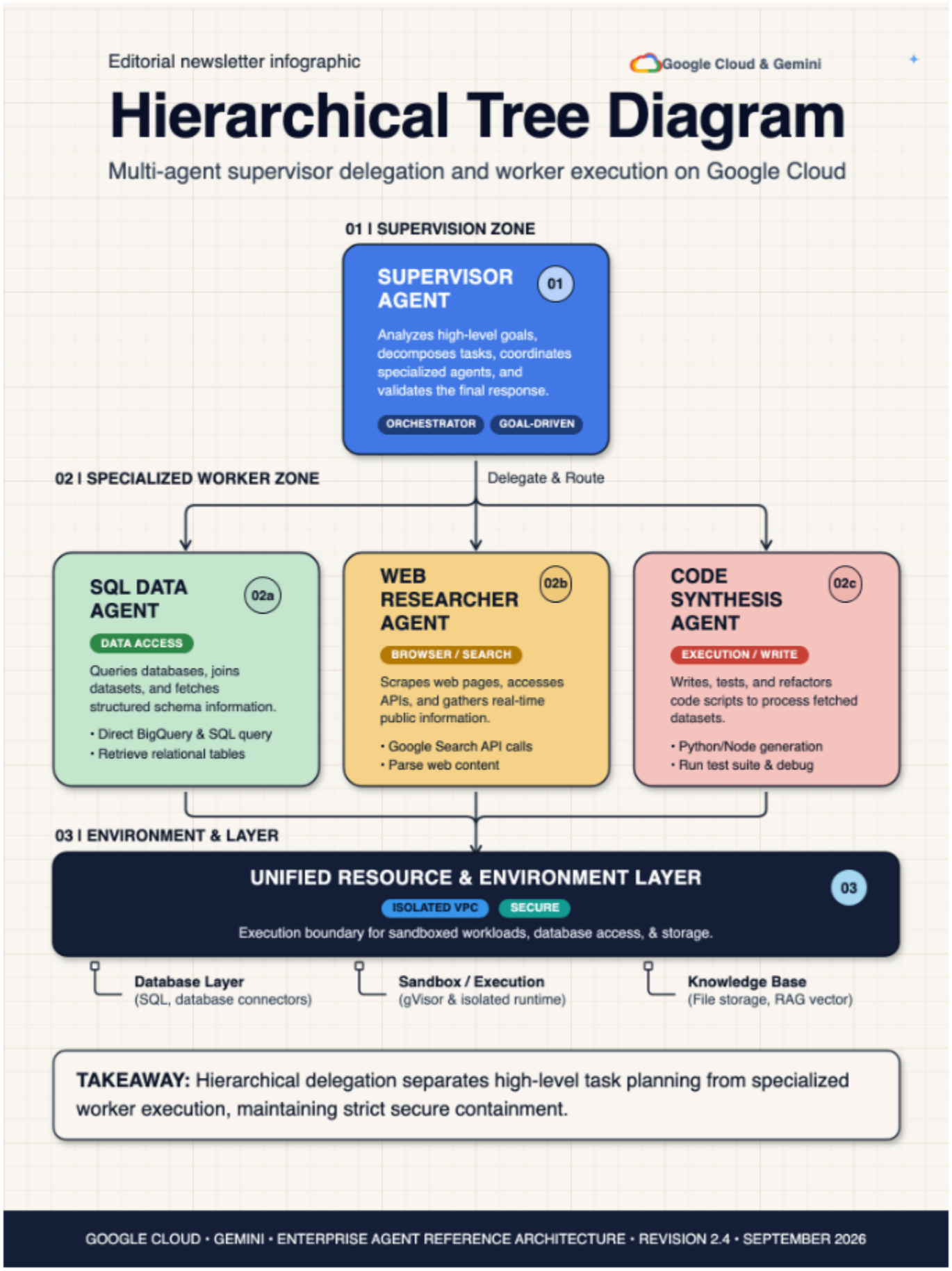
Hierarchical Supervisor Delegation with 5-Stage Progressive Context Compression (1M → 2K)

Prevents context rot across multi-agent A2A workflows by isolating noisy worker tool traces inside sub-sessions and returning only schema-validated artifacts to the root supervisor.



- **Engineering Pillar: Multi-Agent Orchestration, GraphRAG & Memory**
- **Verified Impact: 1,000,000 → 2,000 precision tokens • -78% multi-hop hallucination rate • 3.1x lower cost**
- **Dual Stack: Vertex AI Gemini (locations/global) • Cloud Run ADK • Model Armor**

Why Naive Agent Loops Fail in Enterprise Production

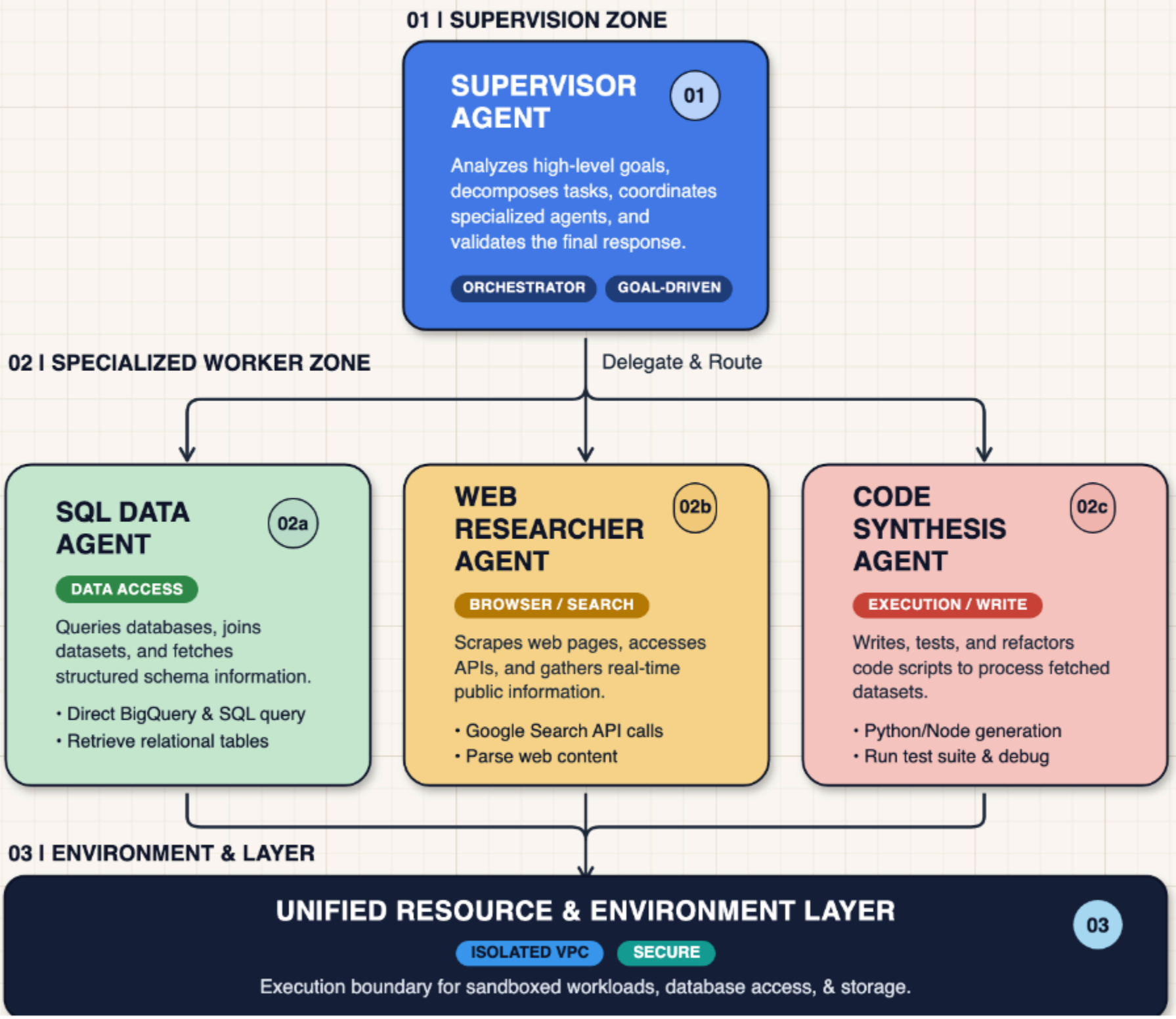
Unbounded context dumps, ambient static API keys, and unverified self-approvals compound latency, cost, and security risk.

Editorial newsletter infographic

Google Cloud & Gemini

Hierarchical Tree Diagram

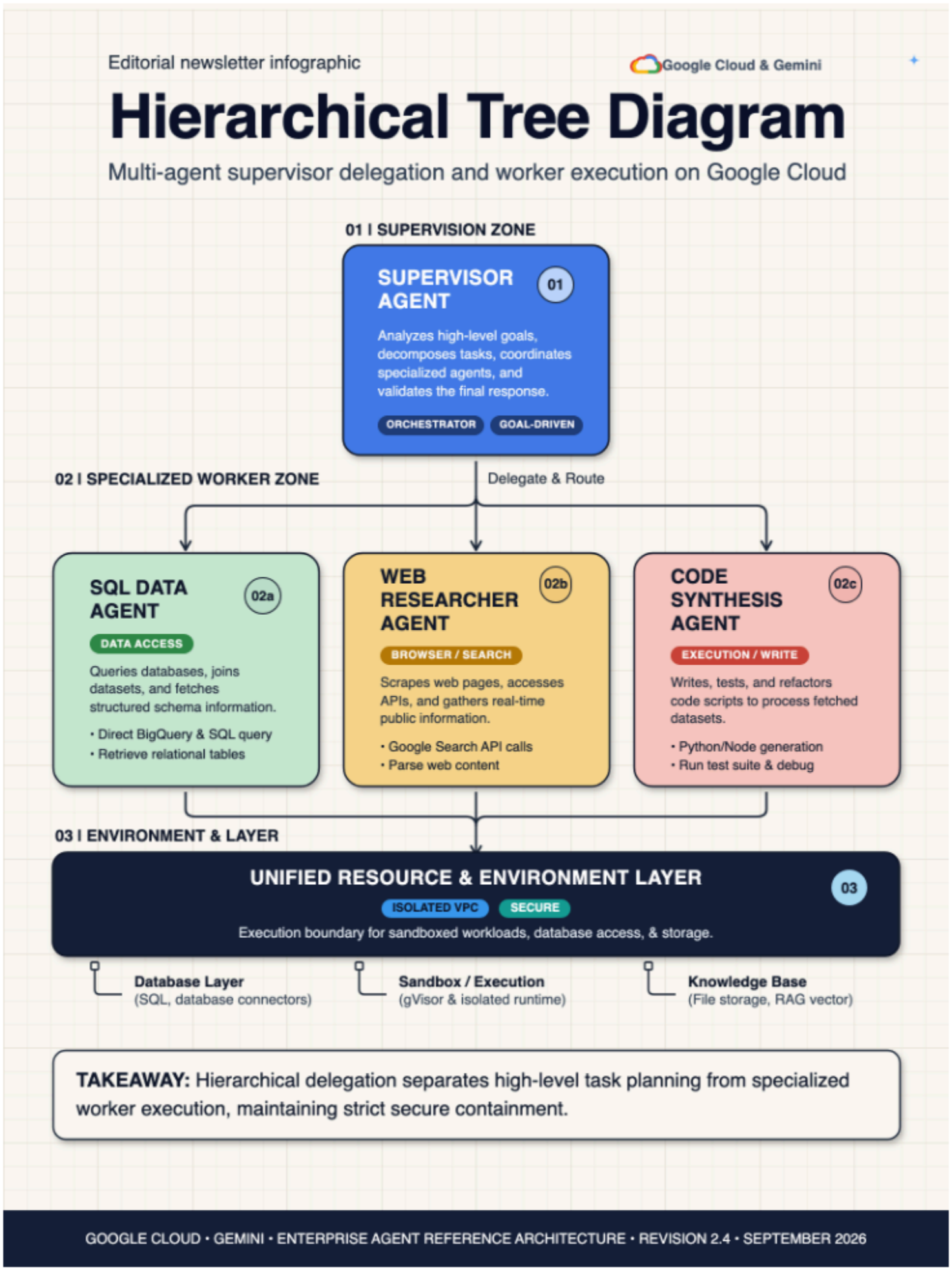
Multi-agent supervisor delegation and worker execution on Google Cloud



- ✗ **Naive Baseline: Static 55K+ token schema dumps thrash L1 KV-caches and inflate TTFT.**
- ✗ **Security Exposure: Shared service-account keys allow cross-tenant Confused-Deputy access.**
- ✓ **Engineered Fix: Enforce deterministic schema budgets, narrow-audience OBO tokens, and CI replay gates outside the LLM.**

Full-Stack Reference Architecture (T62)

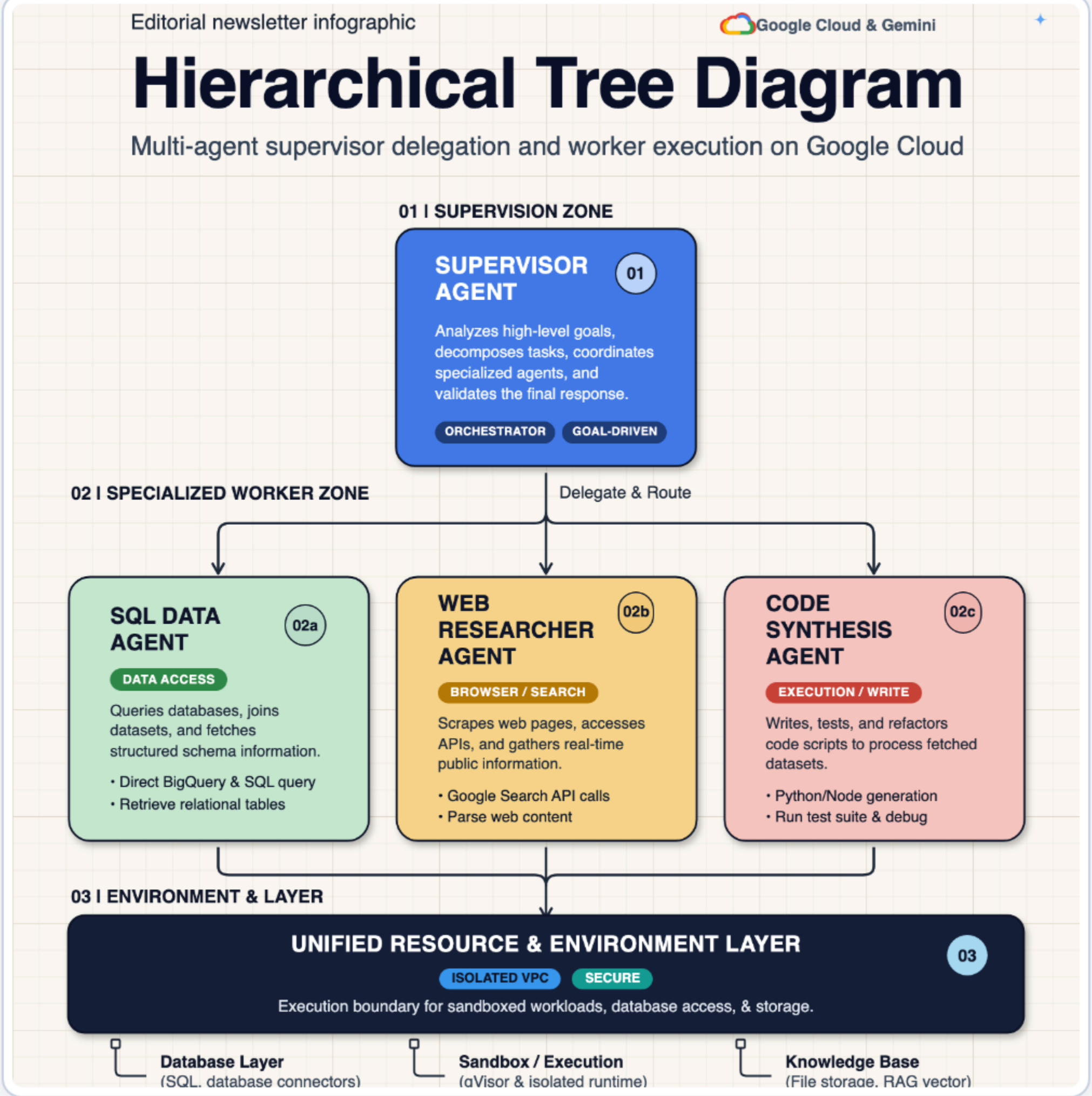
Grounded on Multi-Agent Systems • Context Engineering — pairing managed Google Cloud SLA controls with portable open-source equivalents.



- **Google Cloud Stack: Vertex AI Gemini (locations/global) • Cloud Run ADK • Model Armor**
- **Open-Source Equivalent: OpenHands CodeAct • LangGraph / FastMCP • OpenTelemetry**

Stage 01 — Ingress, Discovery & Context Hygiene

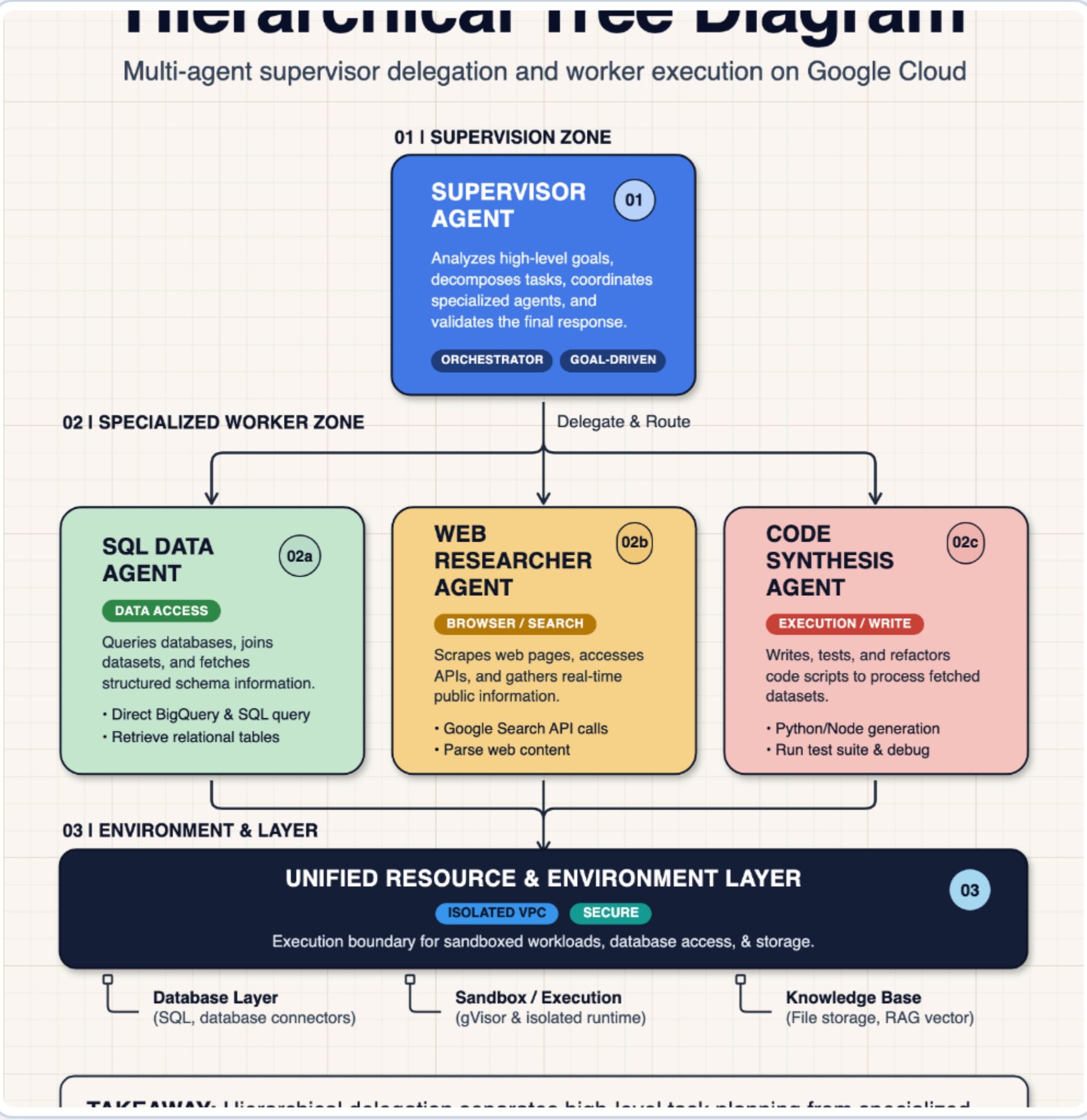
Tier 01 — Scoped Discovery & Context Budget: Mount only Top-K compact tool schemas per turn to preserve KV-cache hits.



- **Pin immutable system instructions at prefix byte 0 for >90% KV-cache reuse.**
- **Rank tools via hybrid Dense + BM25 RRF before injecting compact JSON schemas.**

Stage 02 — Identity, Guardrails & Policy Enforcement

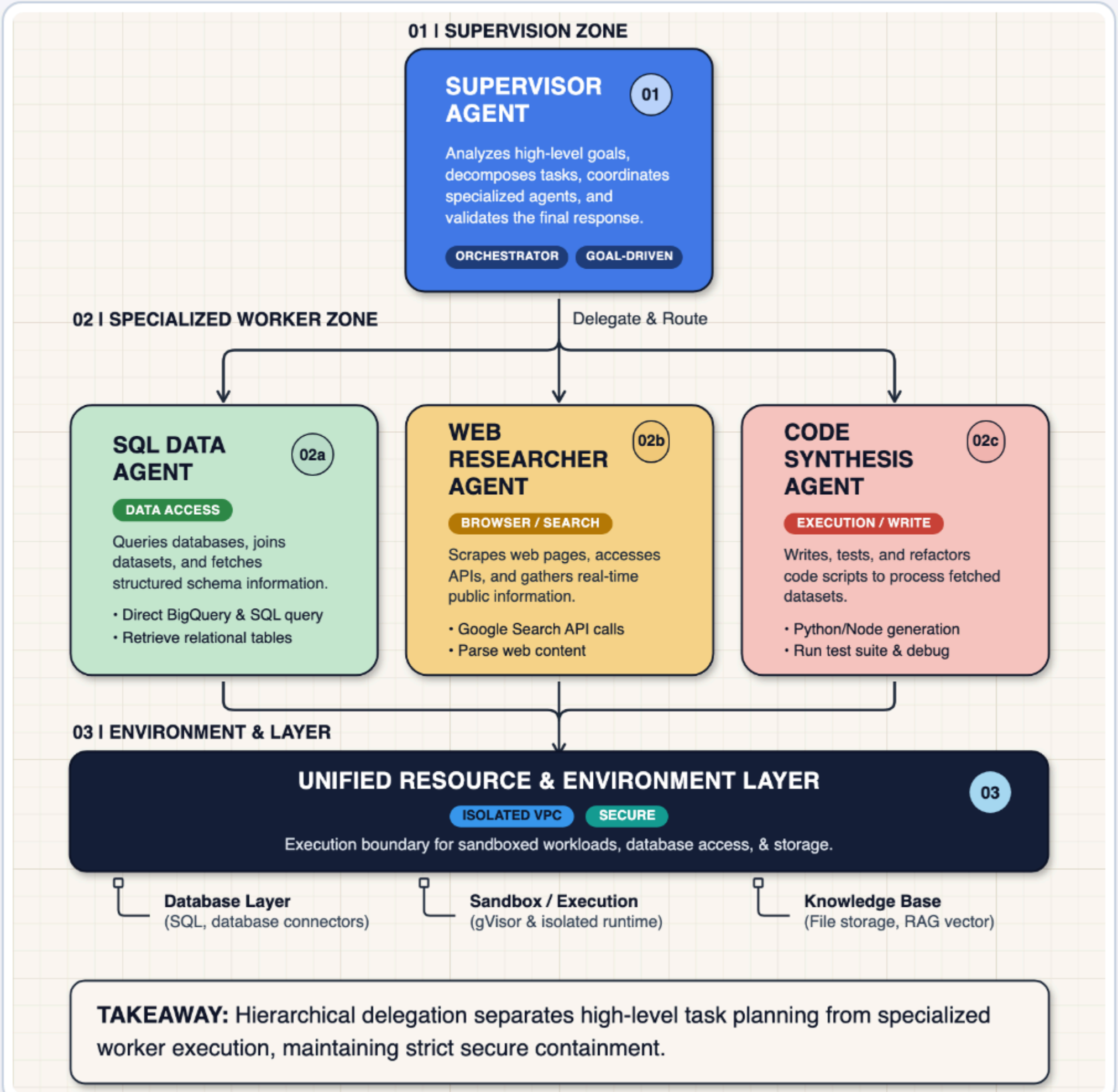
Tier 02 — Zero-Trust Identity & Policy Gate: Exchange caller OIDC tokens via RFC 8693 STS with narrow audience claims.



- **Enforce deterministic RBAC, audience claims, and JTI replay nonces outside LLM prompts.**
- **Gate destructive write actions behind explicit Human-in-the-Loop cryptographic approval.**

Stage 03 — Sandboxed Execution & Adversarial Verification

Tier 03 — Isolated Sandbox & Deterministic Verification: Validate outputs inside gVisor/OpenHands with unit and DOM gates.

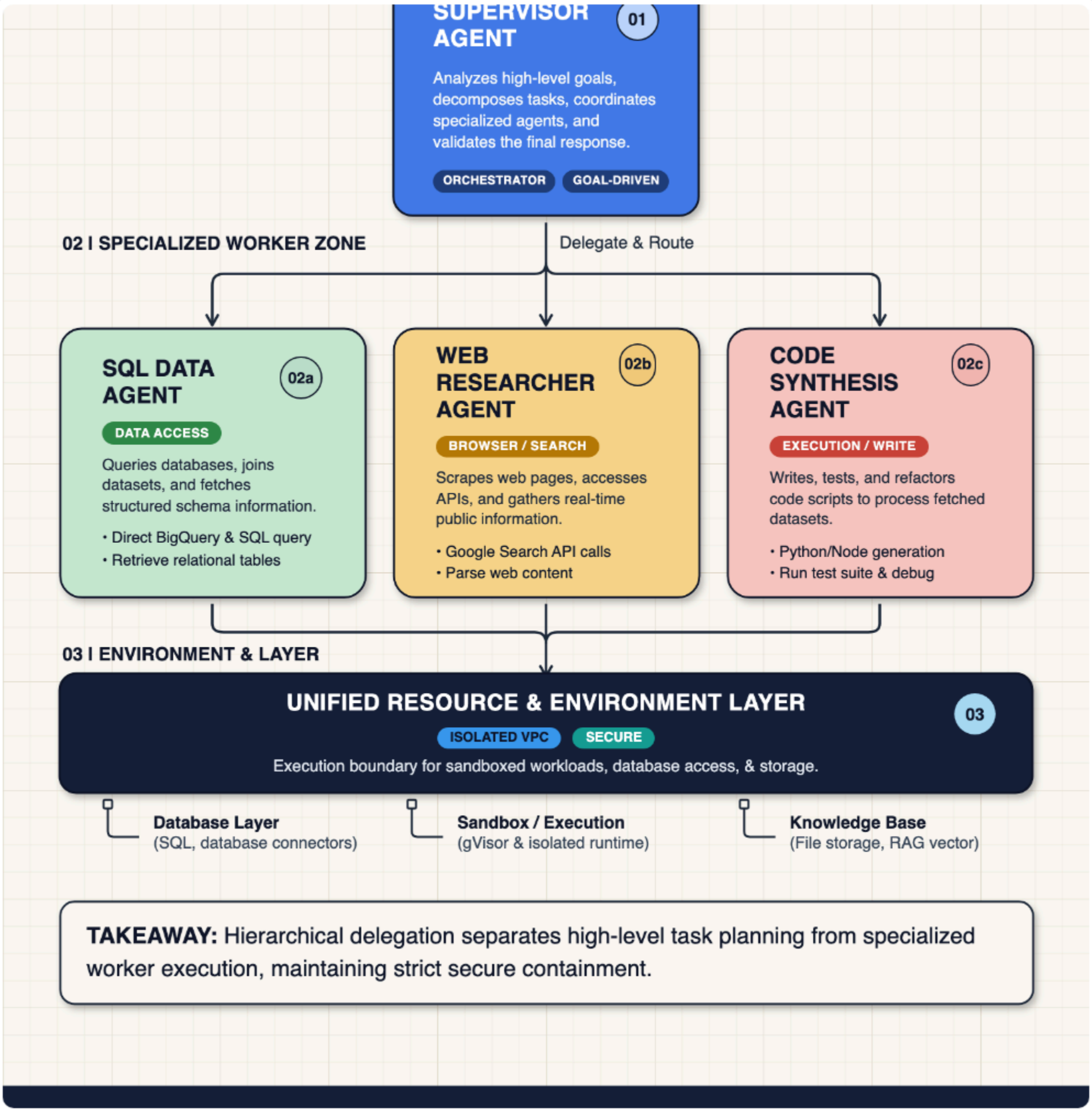


TAKEAWAY: Hierarchical delegation separates high-level task planning from specialized worker execution, maintaining strict secure containment.

- **Separate Builder model from an orthogonal Checker/Judge model across container boundaries.**
- **Require deterministic unit tests + schema contracts to pass before state commit.**

Stage 04 — Production Telemetry, SLOs & FinOps ROI

Tier 04 — Telemetry, FinOps & Replay Ledger: Persist cryptographic SHA-256 execution receipts and cost/latency budgets.

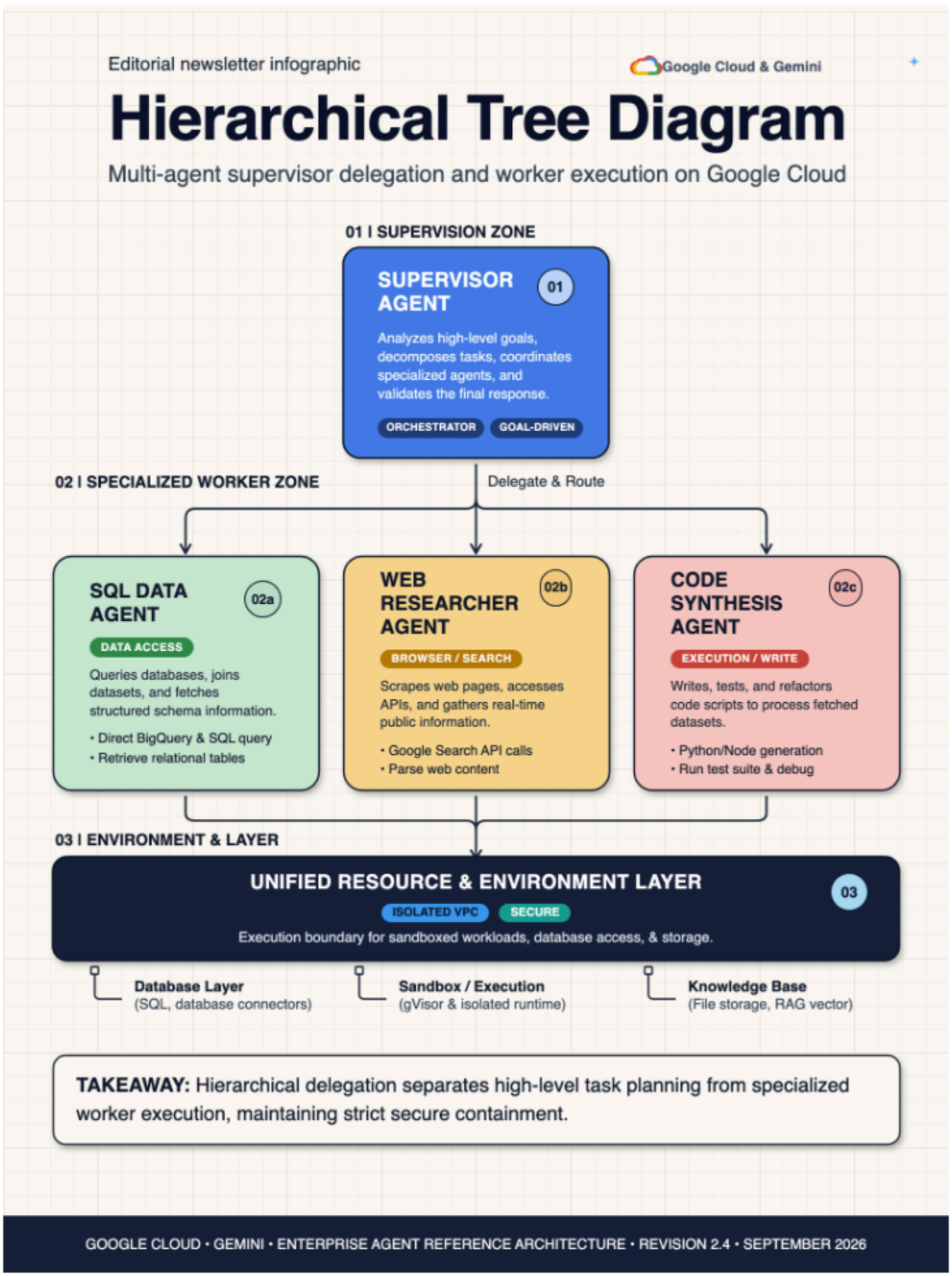


TAKEAWAY: Hierarchical delegation separates high-level task planning from specialized worker execution, maintaining strict secure containment.

- **Measured Telemetry: 1,000,000 → 2,000 precision tokens • -78% multi-hop hallucination rate • 3.1x lower cost**
- **Track p95 end-to-end latency, pass^k multi-trial stability, and cost per verified task.**

Principal Architect Golden Rule (T62)

Enforce deterministic schema budgets, narrow-audience OBO tokens, and CI replay gates outside the LLM.



- **1. Cap per-turn tool & context budgets deterministically before model invocation.**
- **2. Bind narrow-scope OAuth 2.1 OBO identity + Model Armor policy gates.**
- **3. Gate every deployment on a seeded multi-trial replay benchmark.**